PharmacoEconomics (2025) 43:297–310
https://doi.org/10.1007/s40273-024-01457-w

ORIGINAL RESEARCH ARTICLE

A Multistate Model Incorporating Relative Survival Extrapolation and Mixed Time Scales for Health Technology Assessment

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Accepted: 6 November 2024 / Published online: 25 November 2024
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Abstract

Background Multistate models have been widely applied in health technology assessment. However, extrapolating survival in a multistate model setting presents challenges in terms of precision and bias. In this article, we develop an individual-level continuous-time multistate model that integrates relative survival extrapolation and mixed time scales.

Methods We illustrate our proposed model using an illness–death model. We model the transition rates using flexible parametric models. We update the *hesim* package and the *microsimulation* package in R to simulate event times from models with mixed time scales. This feature allows us to incorporate relative survival extrapolation in a multistate setting. We compare several multistate settings with different parametric models (standard vs. flexible parametric models), and survival frameworks (all-cause vs. relative survival framework) using a previous clinical trial as an illustrative example.

Results Our proposed approach allows relative survival extrapolation to be carried out in a multistate model. In the example case study, the results agreed better with the observed data than did the commonly applied approach using standard parametric models within an all-cause survival framework.

Conclusions We introduce a multistate model that uses flexible parametric models and integrates relative survival extrapolation with mixed time scales. It provides an alternative to combine short-term trial data with long-term external data within a multistate model context in health technology assessment.

Key Points for Decision Makers

This study presents a novel individual-level continuous-time multistate model that integrates relative survival extrapolation and mixed time scales, addressing challenges in extrapolating survival data within multistate modeling in health technology assessment.

By incorporating projected general population mortality rates, the model combines short-term clinical trial data and long-term external data, providing an alternative framework for conducting cost-effectiveness analyses.

The proposed model has been implemented using the microsimulation and hesim packages in R, and the authors provide the necessary computer code for users.

1 Introduction

Health economic modeling is widely used in health technology assessment (HTA) to appraise the cost effectiveness of health technologies, including drugs, medical devices, and diagnostic tests. To evaluate the long-term impact of a technology on the general health system, decision makers rely on the estimates produced by the modeling process. This process combines clinical and economic evidence to help evaluate the uncertainties involved in the decision-making [1].

Multistate models have become increasingly popular in HTA. They offer a more realistic representation of the clinical setting that involves time-dependent outcomes [2], making them more suitable than decision trees. These models are built upon a set of mutually exclusive and jointly exhaustive health states that represent various health conditions or treatment options. Typically, there are two main ways to handle time in multistate models: the clock-forward and clock-reset approaches [3]. The clock-forward approach treats time as the time since an individual entered the initial state. This approach holds the Markov property, where transition probabilities to the

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future state can only depend on the initial state and not on the history of any intermediate states [4]. In contrast, the clock-reset approach, referred to as semi-Markov, resets the time to zero when an individual transitions to a new state, allowing the model to account for the time spent in intermediate states [3].

Markov and semi-Markov models can be structured in either continuous or discrete time, at the cohort or individual level, with transition probabilities that may vary over time (time inhomogeneous) or remain constant (time homogeneous) [1, 5, 6]. In health economics, discrete-time Markov cohort models are commonly applied in cost-effectiveness analysis. Discrete-time models may inadequately account for competing risks, even though it is possible to incorporate time-varying risks of death with respect to increasing age. Typically, these models often assume the Markov property, which follows a clock-forward approach [3]. Although they can incorporate memory by including "tunnel states" that track the time spent in a state, this may result in a substantial increase in the number of states (state explosion) [2]. Instead, semi-Markov models adopt a clock-reset approach, relaxing the memoryless assumption by allowing transitions to depend on the time since entering an intermediate state.

Among the various types of multistate models, we focus on individual-level continuous-time multistate models, which offer several advantages. Continuous-time models facilitate the calculation of transition probabilities in the presence of competing risks [7]. The individual-based framework is particularly beneficial for predicting transition probabilities from fitted time-to-event parametric models that involve mixed time scales and are thereby not restricted by the Markov property [2]. For example, models may have transitions to death due to other causes, which are commonly related to attained age and attained calendar period, combined with transitions due to disease progression, which depend on the time since diagnosis or onset of the disease [8]. In circumstances where multiple or mixed time scales are involved, individual-based continuous-time multistate models have been employed in both biostatistics [8–11] and health economics [5, 12–14].

Estimates of long-term survival and costs are frequently required in HTA when the time horizon for an analysis is specified beyond the observed period [15]. In scenarios such as clinical trials, where only limited follow-up time is available, such analyses rely heavily on survival extrapolation. Previous studies have demonstrated that predictions for life-years using relative survival extrapolation are robust among patients with cancer [16–21]. To extrapolate survival within a relative survival framework (RSF) [16], also referred to as an excess hazard framework [18], one typically incorporates general population mortality rates to partition all-cause hazard (mortality) into expected hazard and excess hazard due to the disease of interest. A primary advantage of the RSF is its ability to extrapolate survival based on long-term predictions of general population mortality rates to avoid implausible projections [18]. In this study, we aim to extend this methodology into a multistate model setting. Several studies have investigated the integration of relative survival into multistate modeling, including non-parametric [22], semi-parametric [23–25], and parametric [8] approaches. Of these, we focus on parametric approaches [26], since cost or survival outcomes necessitate extrapolations beyond follow-up. Furthermore, probabilistic sensitivity analysis (PSA) is commonly required in cost-effectiveness analysis to account for uncertainty in parameters by drawing random samples with Monte Carlo simulations [27].

In this article, we develop a multistate model that incorporates relative survival extrapolation and mixed time scales for health economic modeling. Specifically, our focus lies solely on individual-level continuous-time multistate models [5]. In an illustrative example, we explore models using standard parametric models (SPMs) or flexible parametric models (FPMs) using an all-cause survival framework (ASF) or an RSF. We replicate the cost-effectiveness study by Williams et al. [28] using our proposed method as an alternative. Furthermore, to enhance the computational speed of the individual-based continuous-time multistate models, we implement the program in R using computations in C++.

2 Methods

2.1 Multistate Model Incorporating Relative Survival Extrapolation

We begin by illustrating the illness–death model (Fig. 1). An illness–death model is one special type of multistate model for describing disease progression [29]. Typically, the model consists of three states: progression free, progression, and death. For disease onset, the states would represent disease free, illness, and death, respectively. The progression-free state is the initial state where the disease is present but without any progression (illness). The progression state corresponds to the advancement of the disease. The death state represents an absorbing state where no further transitions are possible. We applied a semi-Markov (clock-reset) approach; hence, the choice of time scale for the progression state to the death state is time since progression ($t - u$). A Markov (clock-forward) approach is also available if the choice of time scale is t, time since study entry. Patients may recover from the progression state back to the progression-free state, which is also referred as a "reversible" illness–death model

[30]. For illustrative purposes, we hereby only consider an irreversible model.

Figure 2 shows our suggested extension to the illness–death model. We partition the death state into two states: excess death and expected death. That is, when transitioning to the death state, patients may experience mortality related either to the disease itself (excess death) or to other causes (expected death). Hence, for our modeling application, we have four states (progression free, progression, excess death, and expected death), with five transitions in total. All patients start from the progression-free state and may transition to the progression state or the death state. We also applied a semi-Markov approach for the transitions from the progression state to the excess or expected death state. Below, we describe each transition and its statistical properties individually.

2.1.1 Transition 1: Modeling the Incidence Rate of Progression Free to Progression

All patients start from the progression-free state and thus are at risk for transitioning to the progression state and the excess or expected death states. We use the trial data for the transition from the progression-free state to the progression state to model the transition rate of progression free → progression, $h_1(t)$, with an FPM on a log cumulative hazard scale:

$$\log H_1\left(t|\mathbf{x}_1\right) = \log H_{1,0}(t) + \boldsymbol{\beta}_1\mathbf{x}_1. \quad (1)$$

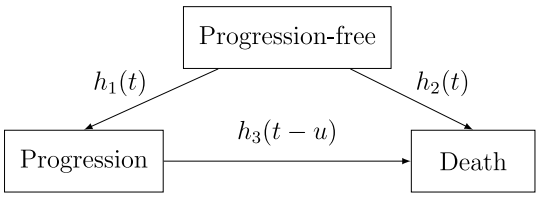
The log cumulative baseline hazard, $\log H_{1,0}(t)$, is modeled with restricted cubic splines [31, 32], and we have the covariate vector $\mathbf{x}_1$, variables of patient characteristics, with the corresponding coefficients $\boldsymbol{\beta}_1$.

2.1.2 Transition 2: Modeling the Mortality Rate of Progression-Free to Death

In an RSF, the all-cause mortality of progression free → death, $h_2(t, t+a, t+c)$, can be written as the sum of the expected mortality, $h_2^*(t+a, t+c)$, and the excess mortality, $\lambda_2(t)$:

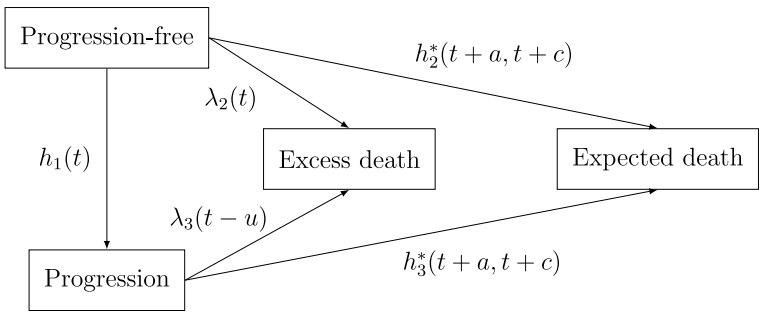
$$h_2\left(t, t+a, t+c|\mathbf{x}_2\right) = h_2^*\left(t+a, t+c|\mathbf{x}_2{}'\right) + \lambda_2\left(t|\mathbf{x}_2\right), \quad (2)$$

where $\mathbf{x}_2$ is a covariate vector that includes patient characteristics such as treatment, age, sex, and calendar year at diagnosis. The expected mortality is obtained from general population mortality rates, which can vary across stratification factors such as age, sex, and calendar year, denoted by $\mathbf{x}_2{}'$, which is a subset of $\mathbf{x}_2$. Here, we use attained age $(t+a)$ and attained calendar year $(t+c)$ as time scales. The extrapolation of the expected mortality for a given birth cohort can be based on projected general population mortality rates, typically published by national vital statistics offices, or by assuming that the rates remain the same as in the last follow-up year [17].



Time scales:
t: Time since study entry
u: Time at progression

Fig. 1 An irreversible illness–death model with three transitions: h_1(t), progression free → progression; h_2(t), progression free → death, and $h_3(t-u)$, progression → death, with a semi-Markov approach



Time scales:
t: Time since study entry
a: Age at study entry
c: Calendar year at study entry
u: Time at progression

Fig. 2 An irreversible illness–death model incorporating a relative survival framework, with five transitions: $h_1(t)$, progression free → progression; $\lambda_2(t)$, progression free → excess death; $h_2^*(t+a, t+c)$, progression free → expected death; $\lambda_3(t-u)$, progression → excess death; and $h_3^*(t+a, t+c)$, progression → expected death, with a semi-Markov approach

Using the trial data in conjunction with the general population mortality rates, any parametric relative survival model can be applied to model the excess mortality of progression free → death, $\lambda_2(t)$. We chose to model this with a flexible parametric relative survival model [33] on a log cumulative excess hazard scale, $\log\Lambda_2\left(t|\mathbf{x}_2\right)$, with time since study entry, t, as the time scale:

$$\log\Lambda_2\left(t|\mathbf{x}_2\right) = \log\Lambda_{2,0}(t) + \boldsymbol{\beta}_2\mathbf{x}_2, \tag{3}$$

where $\log\Lambda_{2,0}(t)$ is the log cumulative baseline hazard, $\mathbf{x}_2$ is the covariate vector, and $\boldsymbol{\beta}_2$ denotes the coefficients.

The all-cause cumulative mortality, $H_2(t, t+a, t+c)$, can be shown as the sum of the cumulative expected mortality, $H_2^*(t+a, t+c)$, and the cumulative excess mortality, $\Lambda_2(t)$, as follows:

$$H_2\left(t, t+a, t+c|\mathbf{x}_2\right) = H_2^*\left(t+a, t+c|\mathbf{x}_2{}'\right) + \Lambda_2\left(t|\mathbf{x}_2\right). \tag{4}$$

Note that model fitting for $H_2(t, t+a, t+c)$ uses the all-cause mortality rates $h_2^*(t+a, t+c)$ for the event times and does not require all-cause cumulative mortality rates $H_2^*(t+a, t+c)$, as minus $H_2^*(t+a, t+c)$ is a constant term in the log-likelihood.

2.1.3 Transition 3: Modeling the Mortality Rate of Progression to Death

Once patients enter the progression state, they are only at risk of transitioning to the death state. We apply a semi-Markov model here, that is, the clock-reset approach [7]; thus, the choice of time scale here is time since progression $(t-u)$ where u is time at progression. The all-cause mortality of progression → death, $h_3(t-u, t+a, t+c)$, can be expressed as the sum of the expected mortality, $h_3^*(t+a, t+c)$, and the excess mortality, $\lambda_3(t-u)$:

$$h_3\left(t-u, t+a, t+c|\mathbf{x}_3\right) = h_3^*\left(t+a, t+c|\mathbf{x}_3{}'\right) + \lambda_3\left(t-u|\mathbf{x}_3\right), \tag{5}$$

where $\mathbf{x}_3$ is a covariate vector containing patient characteristics, and $\mathbf{x}_3{}'$, a subset of $\mathbf{x}_3$, is stratification factors for general population mortality rates. Like transition 2, the expected mortality $h_3^*(t+a, t+c)$ is estimated and extrapolated using attained age $(t+a)$ and attained calendar year $(t+c)$ as time scales, based on data from the general population mortality rates. For the excess mortality, using the trial data for the transition from the progression state to the death state, we chose to model this transition on a log cumulative excess hazard scale, $\log\Lambda_3\left(t-u|\mathbf{x}_3\right)$, with a flexible parametric relative survival model [33]:

$$\log\Lambda_3\left(t-u|\mathbf{x}_3\right) = \log\Lambda_{3,0}(t-u) + \boldsymbol{\beta}_3\mathbf{x}_3, \tag{6}$$

where $\log\Lambda_{3,0}(t-u)$ is the log cumulative baseline hazard, $\mathbf{x}_3$ is the covariate vector with corresponding coefficients $\boldsymbol{\beta}_3$, and $(t-u)$ denotes time since progression as the time scale.

2.1.4 Simulation Algorithm

We focus on using simulation approaches to obtain point estimates of interest in a given multistate model, applying the inversion method [34] to generate random survival times from fitted parametric time-to-event models. Bender et al. [34] proposed this method for a range of parametric distributions, such as exponential, Weibull, and Gompertz. Crowther and Lambert [35] extended this method by incorporating more complex hazard functions, for example, FPMs. In this study, to perform relative survival extrapolation, we use the inversion method to simulate survival times from a piecewise-constant exponential distribution [36], which has been implemented in msm::rpexp R function [37] and the Rpexp C++ class in the microsimulation package [38], within a multistate framework.

We simulate individual trajectories through the transition models until all individuals either reach an absorbing state or are right censored at a specified time, such as the study endpoint. This process treats the multistate model as a series of competing risks [39].

Following Crowther and Lambert [10], the simulation of an individual's trajectory through a multistate model is conducted as follows:

Let a be the initial state entered at time t_a, N_a be the number of allowed transitions from state a, and S be the set of states that can be reached from state a. The study endpoint is denoted as $t_{\max}$.

1. For each state $s \in S$, let $H_{a\to s}(t)$ represent the cumulative hazard function for the transition from state a to state s, where s can be $1, \ldots, N_a$. If $S = \varnothing$, meaning the individual reaches an absorbing state, then the simulation stops.
2. Simulated survival times $\mathrm{T} = \left\{t_{a\to 1}, t_{a\to 2}, \ldots, t_{a\to N_a}, t_{\max}\right\}$ are generated using the inversion method [34] for each allowed transition.
3. The transition with the minimum time from T, denoted $t_{a\to u}$, is selected, and the individual moves to state u at time $t_{a\to u}$. If $t_{a\to u} = t_{\max}$, the simulation stops.
4. Set $a = u$ and $t_a = t_{a\to u}$, and repeat steps 1 to 3 until reaching an absorbing state or $t_{a\to u} = t_{\max}$.

This simulation algorithm enables relative survival extrapolation within the multistate model by simulating survival times in a competing risks framework with mixed time scales. For instance, in transition 2 (Sect. 2.1.2), we can derive transition probabilities from the cumulative expected

hazard function, $H_2^*(t+a, t+c|\mathbf{x}_2')$, and the cumulative excess hazard function, $\Lambda_2(t|\mathbf{x}_2)$.

We repeat this algorithm for a sufficiently large number (n) of individuals through the multistate model [27]. Point estimates of interest—such as mean sojourn times, life-years, quality-adjusted life-years (QALYs), and costs—are calculated by averaging across all simulated individuals. Confidence intervals are computed from repeated simulations for m times using a multivariate normal distribution, with the estimated parameter vector as the mean and the associated variance–covariance matrix as the variance [10, 39].

2.2 Illustrative Example

2.2.1 Data

The illustrative example is based on a randomized clinical trial comparing the use of rituximab in combination with fludarabine and cyclophosphamide (RFC) versus fludarabine and cyclophosphamide alone (FC) for the first-line treatment of chronic lymphocytic leukemia, also referred to as the CLL-8 trial [40]. This study formed the main source of clinical evidence used to inform National Institute for Health and Care Excellence Technology Appraisal TA174 [41, 42]. Our analysis was motivated by the study by Williams et al. [28], who analyzed this trial using semi-Markov multistate modeling in R, considering a competing-risks setting. The data were formatted for multistate modeling from the original wide format (each individual has one row with records on all transitions) into long format (each individual has a separate row for each transition where an individual is at risk). In total, 810 individuals were enrolled in this trial during 2003 to 2006. The RFC arm included 403 patients, and the FC arm 407 patients. Among both arms, the median age was 61 years, and the proportion of men was 74%. [40] Among the RFC arm, there were 106 progressions, 23 deaths after progression, and 21 deaths without progression. The FC arm had 148 progressions, 27 deaths after progression, and 26 deaths without progression. When the results of the CLL-8 trial were first reported [40], the maximum follow-up was approximately 4 years. To evaluate survival extrapolation, we predicted survival from 4 to 50 years using all multistate models (see Sect. 2.2.4) and compared these projections with the updated 8-year follow-up data [43].

2.2.2 Population Mortality Data

To integrate relative survival extrapolation into a multistate model, we require population mortality rates of the study population in the CLL-8 trial. The trial was conducted in 190 centers in 11 countries. We used the general population mortality rates of these countries from the Human Mortality Database [44] to construct the population mortality data for this trial. The data were stratified by calendar year, and age, with average mortality rates calculated considering the varying proportions of males and females across these 11 countries. For extrapolation of the population mortality data beyond 2006 (the last year of enrollment in the CLL-8 trial) [40] and for ages beyond 100 years, we applied the mortality rates from 2006 and age 100 years, respectively.

2.2.3 Hazard Functions

We used the multistate-structured data of the CLL-8 trial [40] provided in the study by Williams et al. [28]. The data were obtained by digitizing the overall survival (OS) and progression-free survival curves that were published in conjunction with the CLL-8 trial [40] using the algorithm proposed by Guyot et al. [45]. The survival curves do not contain sensitive personal information; consequently, the reconstructed data derived from the published figure are also devoid of such information.

We compared the fitted parametric models with the observed hazards by transition. For the observed hazard functions, we applied kernel smoothing methods to plot non-parametric observed hazard functions of 4 years. All parametric models were fitted with the observed data (maximum follow-up of 4 years) and extrapolated to 50 years. Transition 3 (progression to death) used a clock-reset approach with time since progression as the time scale. This approach was recommended by Williams et al. [28], who tested and concluded that the Markov property did not hold, making the semi-Markov model more suitable. We fitted all the parametric models with a treatment covariate, a binary variable indicating RFC or FC. SPMs in this article were referred to as the Gompertz, generalized gamma, and Gompertz distributions for transition 1 to 3, respectively, aligned with the base-case analysis of Williams et al. [28]. For FPMs, each transition had six candidate models: two to four degrees of freedom for main effects with or without two to three degrees of freedom for the covariate to allow for non-proportional hazards [46]. We selected the model with the lowest Akaike information criterion (AIC) [47].

We included three alternative formulations of the multistate model, detailed in Sect. 2.2.4. We applied extrapolation within an ASF for both SPMs and FPMs by fitting the models on a log cumulative hazard scale, and we used extrapolation within an RSF only for FPMs, detailed in Sect. 2.1. Below, we illustrate the characteristics of the parametric models by transition.

(A) **Transition 1: progression free to progression**

For SPMs, we fitted the Gompertz model. For FPMs, the selected model included treatment as a covariate

with two degrees of freedom for the main effect without time-dependent effect.

(B) **Transition 2: progression free to death**

For SPMs, we employed the generalized gamma model. In the case of FPMs, our chosen model included treatment as a covariate with two degrees of freedom for the main effect without time-dependent effect. To extrapolate within an ASF, the model was fitted on a log cumulative hazard scale, $\log H_2\left(t|\mathbf{x}_2\right)$, whereas within an RSF, the model was fitted on a log cumulative excess hazard scale, $\log\Lambda_2\left(t|\mathbf{x}_2\right)$. Subsequently, we obtained the all-cause hazard function using the interrelationship provided in Eq. (2).

(C) **Transition 3: progression to death**

For SPMs, we implemented the Gompertz model. For FPMs, our selected model included treatment as a covariate with three degrees of freedom for the main effect without time-dependent effect. For extrapolation within an ASF, the model was fitted on a log cumulative hazard scale, $\log H_3\left(t|\mathbf{x}_3\right)$, whereas within an RSF, the model was fitted on a log cumulative excess hazard scale, $\log\Lambda_3\left(t|\mathbf{x}_3\right)$, provided in Eq. (6). Hence, we derived the all-cause hazard function using the interrelationship in Eq. (5). For this transition from progression to death, we employed a semi-Markov model (clock-reset) with $(t-u)$, time since progression, as the time scale.

2.2.4 Multistate Models

We summarize the characteristics of all three multistate models in Table 1. Our proposed method was a multistate model that incorporates relative survival extrapolation with mixed time scales using FPMs (FPMs, RSF), whereas the model of Williams et al. [28] used SPMs within an ASF (SPMs, ASF). For comparison purposes, we also present results from FPMs within an ASF (FPMs, ASF). Note that the generalized gamma model did not converge for fitting an excess hazard function for transition 2 (progression free to death), so we did not investigate SPMs in an RSF.

Point estimates of the state occupation probabilities were derived from simulating n = 1,000,000 individuals. Life-years of each state were estimated from the probabilities of being in the non-death states: progression free and progression. Furthermore, we presented the extrapolated survival curves from the six multistate models described above (Table 1) along with mean sojourn time/restricted mean survival time (RMST) and survival proportions at 4, 8, 15, and 50 years or lifetime.

2.2.5 Cost-Effectiveness Analysis

To be consistent with Williams et al. [28], the time horizon of the cost-effectiveness analysis was set at 15 years. We presented the base-case analysis and one-way sensitivity analysis of our proposed model using FPMs within an RSF. The discounting rate for both costs and effectiveness was 3.5%, aligned with Williams et al. [28] To compare the PSA between our model and that by Williams et al. [28], we presented the cost-effectiveness plane and the acceptability curve. PSA involved Monte Carlo simulations [27], Beta Pert distributions [48] for costs, and other distribution parameters used by the manufacturer [42]. We simulated $n = 100,000$ individuals for $m = 1000$ Monte Carlo iterations by refitting the FPMs.

2.3 Statistical Software

We conducted all statistical analyses using R version 4.4.1. We plotted the observed hazard functions using the *muhaz* package [49] and the *bshazard* package [50]. All SPMs and FPMs were fitted and post-estimated by the *flexsurv* package

Table 1 Characteristics of each multistate model include parametric model, survival framework, and time scale by transition

	Transition 1: progression free to progression			Transition 2: progression free to death			Transition 3: progression to death		
	Parametric model	Survival framework	Time scale	Parametric model	Survival framework	Time scale	Parametric model	Survival framework	Time scale
SPMs, ASF (Williams et al. [28])	Gompertz	ASF	t	Generalized gamma	ASF	t	Gompertz	ASF	$t-u$
FPMs, ASF	FPM	ASF	t	FPM	ASF	t	FPM	ASF	$t-u$
FPMs, RSF (proposed method)	FPM	ASF	t	FPM	RSF	t	FPM	RSF	$t-u$

ASF all-cause survival framework, *FPM* flexible parametric model, *RSF* relative survival framework, *SPM* standard parametric model, t time since study entry, $(t-u)$ time since progression

[51] and the *rstpm2* package [52]. Multistate models were computed using the simulation algorithm described in Sect. 2.1.4 using the microsimulation package [38]. For the model using SPMs in an ASF, we additionally employed the *mstate* package [7], following the approach used by Williams et al. For our proposed approach using mixed time scales with FPMs using an RSF, we also updated the *hesim* package [5] by adding an option of simulating event times from models with mixed time scales and included the analysis using the *hesim* package [5] to verify the results. Example code for all analyses in R can be found on the webpage: https://github.com/enochytchen/ChenEYT_microsim.

3 Results

3.1 Comparing Hazard Functions

The observed and extrapolated all-cause hazards by treatment across transitions are shown in Fig. 3 (left column: RFC; right column: FC). For transitions to progression, we extrapolated the fitted hazard functions using the Gompertz model and the FPMs, respectively. Note that there was no RSF involved in this transition (transition 1), as the event was progression instead of death. For transitions to death (transitions 2 and 3), we also considered FPMs within an RSF, of which the hazard function was higher than the same FPMs within an ASF, especially in the long run.

3.2 Comparing Survival Extrapolations

The observed and the extrapolated OS from all six multistate models for both arms—RFC (left panel) and FC (right panel)—are presented in Fig. 4. This includes the base-case analysis of the study by Williams et al. [28] and our proposed model using FPMs within an RSF. For the observed data, the black curves are the 4-year Kaplan–Meier survival estimates from the CLL-8 trial [40], and the green curves present the 8-year Kaplan–Meier survival estimates reported by Fischer et al. [43]. All three multistate models were fitted using the initial 4-year CLL-8 trial data and predicted up to 50 years. The 8-year data were included specifically for comparison with the extrapolated survival estimates from the multistate models. Within the ASF, the models using SPMs predicted lower estimates than those using FPMs. Multistate models using FPMs within an ASF or an RSF showed similar estimates for the first 10 years; models within an RSF generated a lower survival curve over the long run (>10 years).

We summarize the observed and extrapolated survival outcomes, including mean sojourn time/RMST and survival proportions, at 4, 8, 15, and 50 years or lifetime for patients treated with FC and RFC in Table 2. These estimates were generated without discounting for comparison purposes.

For the FC arm, the observed 4-year survival proportion of OS was 0.76, decreasing to 0.39 at 8 years, with corresponding RMST of 3.53 years at 4 years and 5.87 years at 8 years. In comparison, the RFC arm had a higher observed 4-year OS survival proportion of 0.83, declining to 0.55 at 8 years, with RMST of 3.66 years at 4 years and 6.44 years at 8 years. When considering extrapolated outcomes, the models using FPMs within an ASF and an RSF both provided predictions that agreed better with the observed RMST for OS at 8 years, with estimates of 5.86 years for FC and 6.12 years for RFC. However, for the 8-year survival proportion of OS, the FPMs within an ASF or an RSF predicted 0.45 for FC, which was higher than the observed value of 0.39, whereas the SPMs, ASF model predicted 0.34, which was closer to the observed.

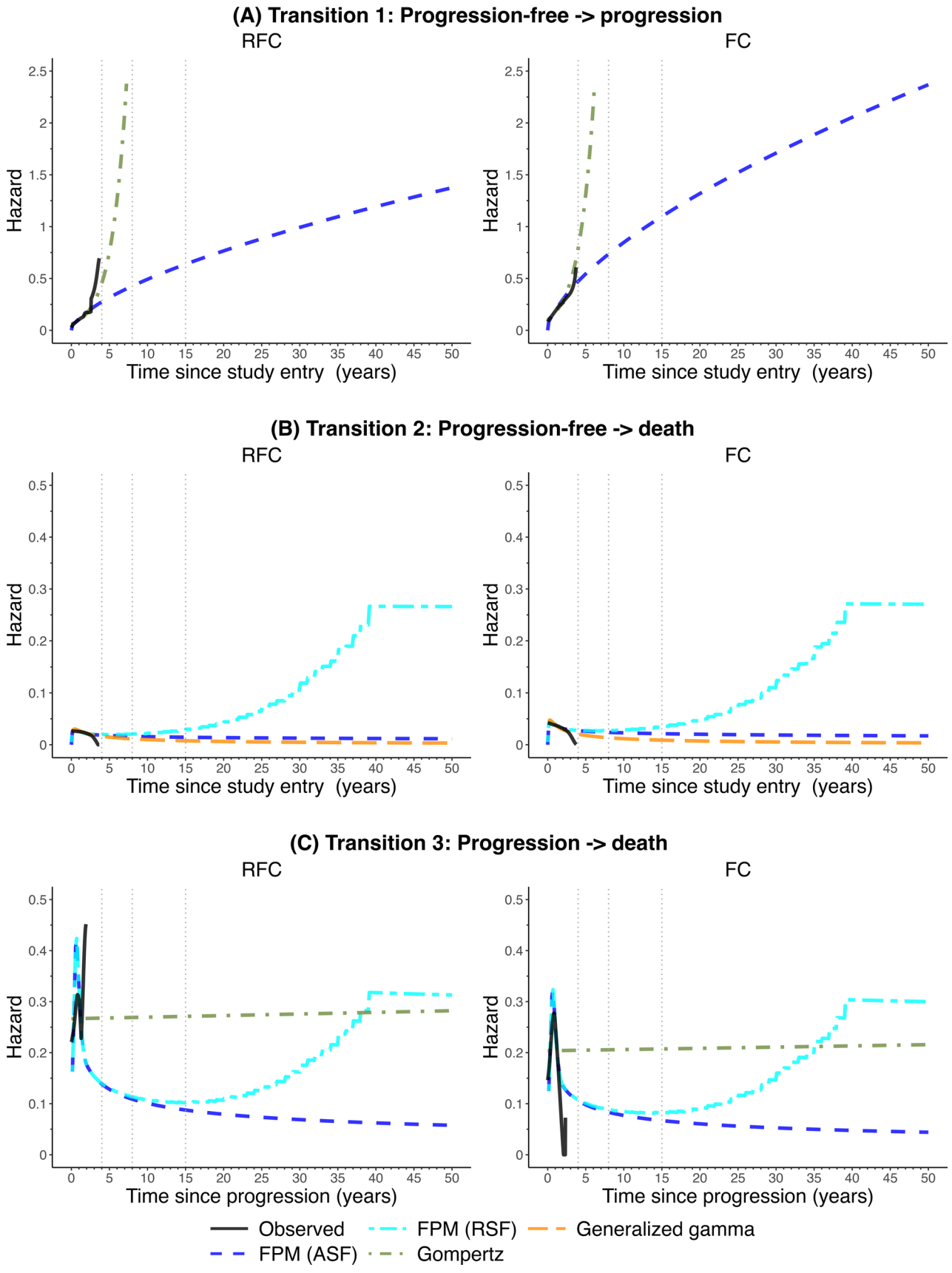
For the progression-free state**,** the observed 8-year survival proportions were 0.10 for FC and 0.27 for RFC, with mean sojourn time of 3.52 and 4.72 years, respectively. All three models overestimated survival proportions and mean sojourn time. The FPMs, ASF model and the FPMs, RSF model had the closest survival proportion estimates (0.02 for FC, 0.11 for RFC) and mean sojourn time (3.00 years for FC, 4.05/4.06 years, respectively, for RFC) to the observed.

For the progression state, the observed 8-year survival proportions were 0.41 for FC and 0.26 for RFC, with mean sojourn time of 2.35 and 1.72 years, respectively. The FPMs, ASF model and the FPMs, RSF model provided the closest estimates for mean sojourn time compared with the observed. All models generally overestimated the survival proportions compared with the observed data, except that the SPMs, ASF model underestimated the survival proportion of 0.34 for FC.

The predicted RMST and survival proportions were similar between the FPMs, ASF model and the FPMs, RSF model at 4, 8, and 15 years. However, over a 50-year period or lifetime, the estimates from the FPMs, RSF model were lower than or equivalent to those from the FPMs, ASF model.

3.3 Comparing Cost-Effectiveness Analyses

Using these multistate models, we extended the cost-effectiveness analyses by Williams et al. [28]. The base-case analysis and one-way sensitivity analyses are presented in supplementary material A and B, respectively. It is important to note that a discounting rate of 3.5% applied in the cost-effectiveness analysis, resulting in discounted mean life-years in supplementary material A that differ from the undiscounted estimates in Table 2. The incremental cost per QALY gained for RFC versus FC was estimated at £34,511


(A) Transition 1: Progression-free -> progression
RFC
FC
Hazard
Time since study entry (years)
(B) Transition 2: Progression-free -> death
RFC
FC
Hazard
Time since study entry (years)
(C) Transition 3: Progression -> death
RFC
FC
Hazard
Time since progression (years)
Observed
FPM (RSF)
Generalized gamma
FPM (ASF)
Gompertz

◂**Fig. 3** Observed and extrapolated all-cause hazard functions by treatment arm for transitions. **a** Transition 1: progression free → progression, **b** Transition 2: progression free → death, and **c** transition 3: progression → death. The observed hazard functions were with maximum 4 years of follow-up; the parametric hazard functions were extrapolated by flexible parametric models (FPMs) within an all-cause survival framework (ASF) or a relative survival framework (RSF), the Gompertz model, and the generalized gamma model. Vertical dashed lines indicate time points at 4, 8, and 15 years. FC, fludarabine and cyclophosphamide; RFC, rituximab, fludarabine and cyclophosphamide

(supplementary material A), which was lower than the £37 665 estimated by Williams et al. [28].

Figure 5 shows the cost-effectiveness planes (Fig. 5a) and the cost-effectiveness acceptability curves (Fig. 5b). To compare, we also presented the PSA results from Willliams et al. [28] in the same figures. The results indicated that, when applying the same cost-effectiveness threshold, for example, £30,000 per QALY gained, the probability of RFC being cost effective was greater using our proposed FPMs, RSF model (50.7%) than with the SPMs, ASF model (36.3%) by Williams et al. [28] (Fig. 5b).

4 Discussion

This study introduced a multistate model that integrates FPMs with relative survival extrapolation and mixed time scales. We demonstrated the implementation using an illness–death model as an example using the *microsimulation* package and the *hesim* package in R. A notable feature of survival extrapolation within an RSF is its capacity to predict survival using long-term general population mortality rates, which helps avoid unrealistic projections [18]. To our knowledge, this is the first method to incorporate FPMs within an RSF into a multistate model, with direct application for cost-effectiveness analysis.

In the illustrative example, our findings showed that the predictions from the FPMs within an ASF or an RSF in general agreed better with the observed data than did the SPMs within an ASF (Fig. 4; Table 2). However, we acknowledge that the extrapolated OS (from any of the three multistate models) for the RFC arm did not agree perfectly with the observed data at 8 years. Insights from domain experts suggest a turning point exists for transition 1 (progression free → progression), meaning that the hazard function may not remain a monotonically increasing function beyond 4 years (Fig. 3). Although Williams et al. [28] applied the Gompertz distribution for modeling this transition, we applied an FPM. The extrapolated hazard functions from both models depended on coefficients derived from the fitted parametric models, which may not have captured the potential turning point in the long run. To improve the prediction, we recognize that alternative models, even those with worse AIC values, could be explored to seek greater plausibility. However, it remains challenging to decide the location of the turning point and the shape of the hazard function before and after the turning point. One potential alternative is to use real-world cancer registry data on chronic lymphocytic leukemia progression, which may help predict the transition from progression free to progression. We suggest that future research explores methods for predicting progression rates more accurately.

In this article, we compared multistate models using SPMs and FPMs within an ASF or an RSF. With the advent of immunotherapies [53] and chimeric antigen receptor T-cell therapies [54], a subset of patients may achieve statistical cure, where their long-term survival becomes similar or equivalent to that of the general population. In such cases, alternative models, such as mixture or non-mixture cure models [55], may be appropriate for modeling transitions to death, such as progression free → death and progression → death in the illness–death model (Fig. 1).

The advancements presented in this article were first inspired by our earlier study, which showed that more accurate survival extrapolation may be carried out by extrapolating the expected mortality and the excess mortality separately [16]. Relative survival extrapolation adopts population mortality rates, which may explain a substantial proportion of the long-term all-cause hazard due to the aging of the patient cohort [18]. In contrast, extrapolation within an ASF does not take this external information into account. In cost-effectiveness analyses, multistate models are widely used as a modeling approach [6, 28, 56]. To integrate relative survival extrapolation in a multistate setting, one needs to handle prediction from mixed time scales because the excess hazard and the expected hazard are estimated on different time scales. As a further contribution, we provide computer code for simulating event times from models with mixed time scales in both the *microsimulation* package and the *hesim* package in R.

Predicting survival involves making assumptions about how trends will continue beyond the observed period. It is crucial to note that relative survival extrapolation also depends on several assumptions. First, we assumed that the average general population mortality rates from 11 countries are the expected mortality of the patients if they had not been diagnosed with the cancer of study. If this assumption does not hold, we need a population mortality rate file stratified on additional factors other than age, sex, and calendar year to reduce the impact of this issue [57, 58]. Second, we assumed that the population mortality rates remain the same as 2006 and extrapolated to the long run. It was shown that using different assumptions for the future general population

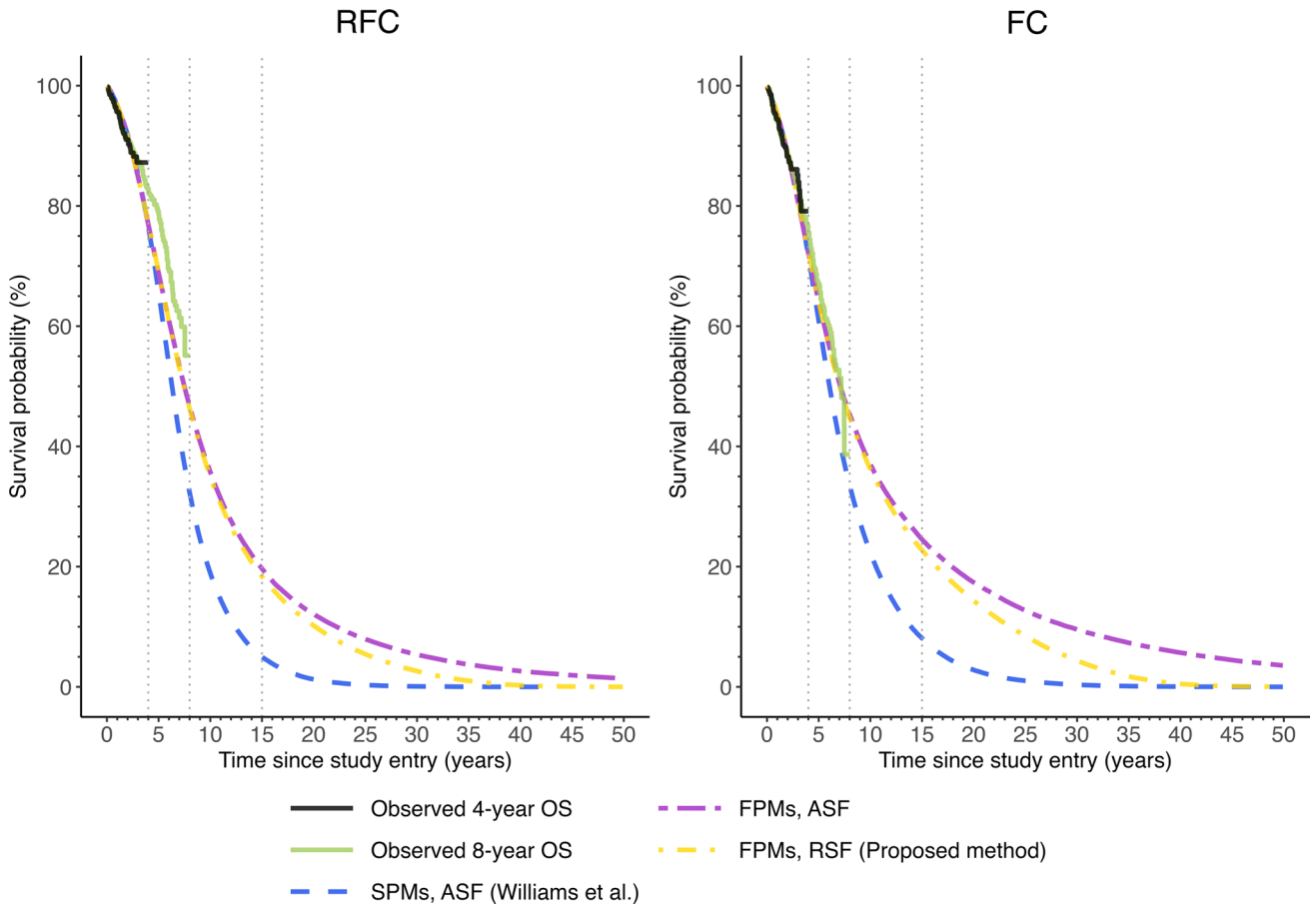


Fig. 4 Observed and extrapolated survival functions for patients of the rituximab, fludarabine and cyclophosphamide (RFC) arm (left) and the fludarabine and cyclophosphamide (FC) arm (right) in the CLL-8 trial. Observed overall survival (OS), estimated with the Kaplan–Meier estimator, is shown by Hallek et al. (observed 4-year OS in black) and Fischer et al. (observed 8-year OS in green). Extrapolated OS for 50 years is shown by the following models: models using standard parametric models (SPMs) within an all-cause survival framework (ASF) and using flexible parametric models (FPMs) within an ASF or a relative survival framework (RSF). The dotted lines indicate 4, 8, and 15 years

mortality rates, such as using projected rates or assuming constant as the last available year, might impact younger ages more than older ages [17]. We recommend conducting sensitivity analysis on different assumptions on the future general population mortality rates. Third, given the lack of information on individual-level patient data, we generated the general population mortality rates by weighting them according to sex proportions and averaging them across 11 countries. We subsequently predicted the expected hazard for a 61-year-old patient in 2005, aligning with the median age of the CLL-8 trial [40] participants in the 2003–2006 enrollment period. We suggest that the relative survival model should adopt the expected mortality rates on an individual level by age, sex, calendar year, and country when such data are available. However, mild mis-specification of the expected mortality rates is expected to have limited impact on the relative survival model predictions [17].

One limitation of this study is that we were only able to demonstrate that survival extrapolation using our proposed multistate model was in general closer to the observed 8-year follow-up than was the base-case model by Williams et al. [28]. However, we were unable to assess the accuracy of our proposed method (FPMs, RSF) for predicting survival beyond 8 years. HTAs are often conducted when patient survival data are heavily right censored. Therefore, we recommend incorporating longer-term observations to improve the evaluation of newly introduced pharmaceutical products. Another limitation is that we only considered an irreversible illness–death model as an illustrative example. However, reversible transitions (progression → progression free), such as recovery or remission, are commonly considered in HTA cases. Our proposed method is adaptable to reversible multistate models as well as to multistate models that include a separate state for those recovered. Special attention should be given to the extrapolation when including

Table 2 Observed and extrapolated survival outcomes: undiscounted mean sojourn time/RMST, and survival proportions, for the CLL-8 trial for the RFC arm and the FC arm at 4, 8, 15, and 50 years or lifetime

			Mean sojourn time/RMST in years at times (years)				Survival proportions at times (years)			
Treatment	State	Source	4	8	15	50*	4	8	15	50*
FC	OS	**Observed**	**3.53**	**5.87**			**0.76**	**0.39**		
		SPMs, ASF (Williams et al. [28])	3.53	5.61	6.87	7.08	0.73	0.34	0.08	0.00
		FPMs, ASF	3.51	5.86	8.16	11.66	0.72	0.45	0.25	0.04
		FPMs, RSF (proposed method)	3.52	5.86	8.08	10.06	0.72	0.45	0.23	0.00
	Progression-free	**Observed**	**2.63**	**3.52**			**0.35**	**0.10**		
		SPMs, ASF	2.55	2.74	2.74	2.74	0.22	0.00	0.00	0.00
		FPMs, ASF	2.56	3.00	3.03	3.03	0.28	0.02	0.00	0.00
		FPMs, RSF	2.56	3.00	3.02	3.02	0.28	0.02	0.00	0.00
	Progression	**Observed**	**0.90**	**2.35**			**0.41**	**0.41**		
		SPMs, ASF	0.99	2.87	4.12	4.33	0.51	0.34	0.08	0.00
		FPMs, ASF	0.95	2.86	5.14	8.64	0.45	0.43	0.25	0.04
		FPMs, RSF	0.96	2.86	5.06	7.04	0.45	0.43	0.23	0.00
RFC	OS	**Observed**	**3.66**	**6.44**			**0.83**	**0.55**		
		SPMs, ASF	3.63	5.80	6.85	6.97	0.77	0.33	0.05	0.00
		FPMs, ASF	3.62	6.12	8.24	10.39	0.77	0.47	0.20	0.01
		FPMs, RSF	3.62	6.12	8.17	9.55	0.77	0.46	0.18	0.00
	Progression-free	**Observed**	**3.15**	**4.72**			**0.57**	**0.27**		
		SPMs, ASF	3.01	3.52	3.52	3.52	0.42	0.00	0.00	0.00
		FPMs, ASF	3.02	4.05	4.25	4.26	0.47	0.11	0.00	0.00
		FPMs, RSF	3.02	4.06	4.25	4.26	0.47	0.11	0.00	0.00
	Progression	**Observed**	**0.51**	**1.72**			**0.26**	**0.26**		
		SPMs, ASF	0.62	2.28	3.33	3.45	0.35	0.33	0.05	0.00
		FPMs, ASF	0.60	2.06	3.99	6.13	0.30	0.36	0.19	0.01
		FPMs, RSF	0.60	2.06	3.92	5.29	0.30	0.36	0.18	0.00

*If the survival proportion at 50 years is zero, the reported 50-year RMST is considered as life expectancy

The observed outcomes were estimated from the Kaplan–Meier survival functions from Hallek et al. [40] and Fischer et al. [43]. The extrapolated outcomes were from the following models: semi-Markov models using SPMs within an ASF and using FPMs within an ASF or an RSF

ASF all-cause survival framework, *FC* fludarabine and cyclophosphamide, *FPMs* flexible parametric models, *LE* life expectancy, *OS* overall survival, *RFC* rituximab, fludarabine, and cyclophosphamide, *RSF* relative survival framework, *RMST* restricted mean survival time, *SPMs* standard parametric models

reversible transitions in multistate modeling. This includes, for example, performing visual assessments on the extrapolated hazard functions and seeking input from clinical or domain experts. Third, the PSA results from our proposed method exhibited greater dispersion than those from the model by Williams et al. [28]. This increased uncertainty can be attributed to the use of FPMs in our approach, which incorporate spline terms and introduce additional complexity compared with the SPMs employed by Williams et al. [28], which used simpler functional forms such as the Gompertz and generalized gamma distributions. Last, in this study, we selected the transition models based solely on AIC. However, we recommend that future model selection incorporates domain knowledge and visual assessment, particularly when long-term extrapolation is involved. These additional approaches offer meaningful insights that complement information criteria, such as AIC, which assess only the goodness of fit within the observed data.

Previous research indicated that life expectancy for patients with cancer can be estimated accurately using relative survival extrapolation, if there is sufficient follow-up data, with at least 10 years recommended [16]. However, cost-effectiveness analyses in randomized clinical trials typically involve follow-up less than 5 years. Extrapolating survival from heavily right-censored data while partitioning survival in a multistate model is indeed a substantial challenge for accurate prediction. Instead, our focus lies in introducing and implementing relative survival extrapolation and

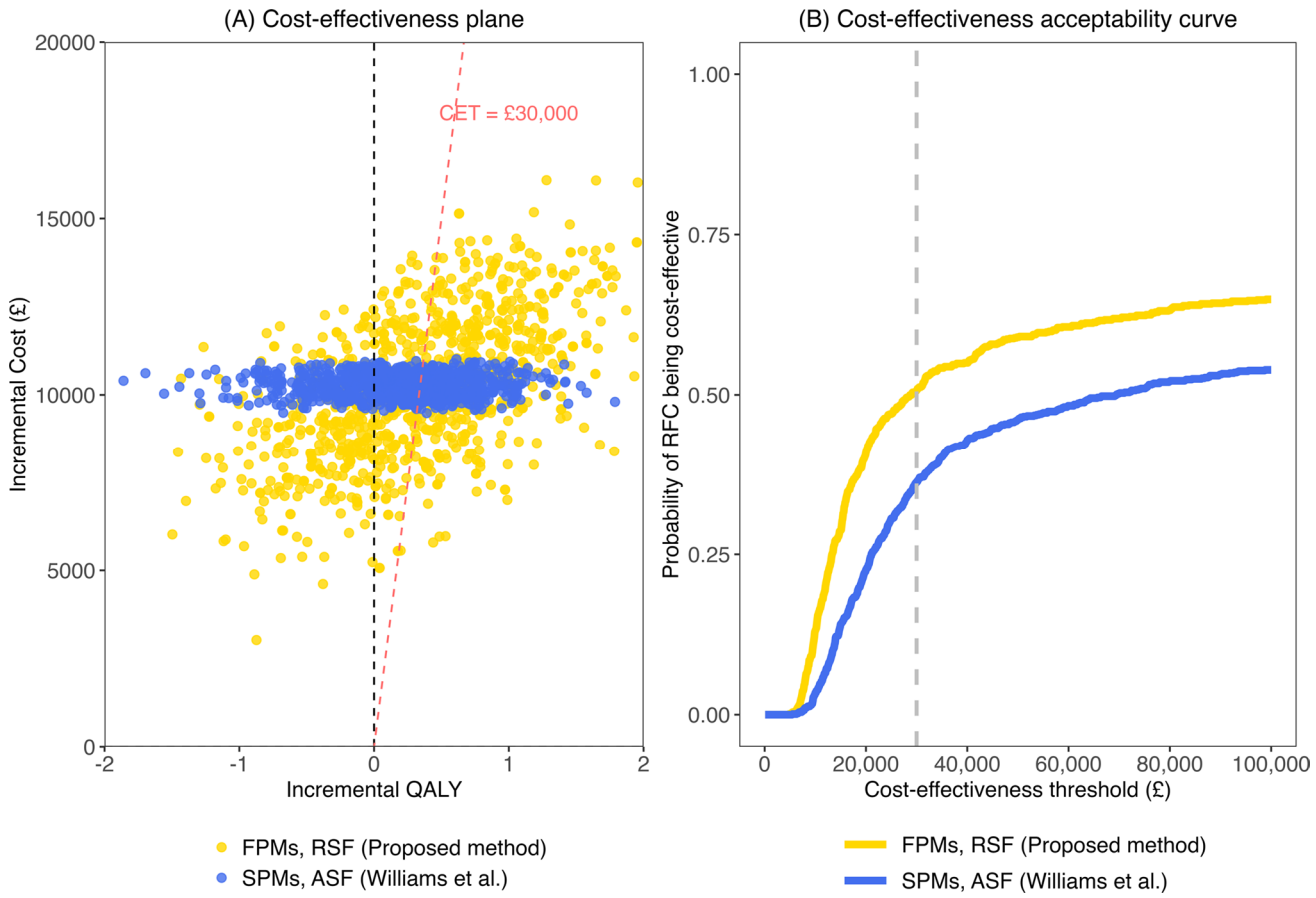


Fig. 5 Cost-effectiveness plane (left) and cost-effectiveness acceptability curve (right). The dashed line for the right panel indicates £30,000. *ASF* all-cause survival framework, *CET* cost-effectiveness threshold, *FPM* flexible parametric model, *QALY* quality-adjusted life-years, *RFC* rituximab, fludarabine and cyclophosphamide, *RSF* relative survival framework, *SPM* standard parametric model

mixed time scales within a multistate model framework. We recommend future research to evaluate this newly proposed method, employing various parametric models to assess its predictive accuracy in a multistate model context.

5 Conclusions

Cost-effectiveness analyses frequently involve right-censored survival data, where survival extrapolation is required. We proposed a multistate model that uses FPMs and incorporates relative survival extrapolation with mixed time scales. This approach integrates projected long-term general population mortality rates into multistate models, enhancing the ability to combine short-term trial data with long-term external data in HTA. This dynamic health economic modeling framework offers an alternative for extrapolating survival data in a multistate model context.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40273-024-01457-w.

Acknowledgements The authors thank Caroline E. Dietrich, PhD, and Michael J. Crowther, PhD, who provided their statistical insight in multistate modelling; Robert Szulkin, PhD, for reviewing the manuscript, and Devin Incerti, PhD, for his guidance on the hesim implementation.

Declarations

Funding Open access funding provided by Karolinska Institute. This research was funded by research grants by the Swedish Research Council and Cancerfonden (The Swedish Cancer Society).

Role of the funders/sponsors The funding agreement ensured the authors' independence in designing the study, interpreting the data, and writing and publishing the report.

Conflict of interest All authors have no relationship to disclose.

Author contributions Concept and design: Enoch Yi-Tung Chen, Paul W. Dickman, Mark S. Clements. Analysis and interpretation of data: Enoch Yi-Tung Chen, Paul W. Dickman, Mark S. Clements. Drafting of the manuscript: Enoch Yi-Tung Chen, Paul W. Dickman, Mark S. Cle-

ments. Critical revision of the paper for important intellectual content: Enoch Yi-Tung Chen, Paul W. Dickman, Mark S. Clements. Statistical Analysis: Enoch Yi-Tung Chen, Mark S. Clements. Obtaining funding: Paul W. Dickman. Supervision: Paul W. Dickman, Mark S. Clements.

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References

1. Briggs AH, Claxton K, Sculpher MJ. Decision modelling for health economic evaluation. Oxford: Oxford University Press; 2006.
2. Siebert U, Alagoz O, Bayoumi AM, Jahn B, Owens DK, Cohen DJ, et al. State-transition modeling: a report of the ISPOR-SMDM modeling good research practices task force-3. Value Health. 2012;15:812–20.
3. Putter H, Fiocco M, Geskus RB. Tutorial in biostatistics: competing risks and multi-state models. Stat Med. 2007;26:2389–430.
4. Davis S, Matt S, Tappenden P, Wailoo A. NICE DSU Technical Support Document 15: cost-effectiveness modelling using patient-level simulation [Internet]. 2014. Available from: http://www.nicedsu.org.uk.
5. Incerti D, Jansen JP. hesim: health economic simulation modeling and decision analysis [Internet]. 2021. Available from: https://arxiv.org/abs/2102.09437.
6. Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW, Drummond MF, et al. Methods for the economic evaluation of health care programmes. 4th ed. Oxford: Oxford University Press; 2015.
7. Putter H, Fiocco M, Geskus RB. Tutorial in biostatistics: competing risks and multi-state models. Statist Med. 2007;26:2389–430.
8. Weibull CE, Lambert PC, Eloranta S, Andersson TML, Dickman PW, Crowther MJ. A multistate model incorporating estimation of excess hazards and multiple time scales. Stat Med. 2021;40:2139–54.
9. Iacobelli S, Carstensen B. Multiple time scales in multi-state models. Statist Med. 2013;32:5315–27.
10. Crowther MJ, Lambert PC. Parametric multistate survival models: flexible modelling allowing transition-specific distributions with application to estimating clinically useful measures of effect differences. Stat Med. 2017;36:4719–42.
11. Fiocco M, Putter H, van Houwelingen HC. Reduced-rank proportional hazards regression and simulation-based prediction for multi-state models. Statist Med. 2008;27:4340–58.
12. Karlsson A, Jauhiainen A, Gulati R, Eklund M, Grönberg H, Etzioni R, et al. A natural history model for planning prostate cancer testing: Calibration and validation using Swedish registry data. Shore N, editor. PLoS ONE. 2019;14:e0211918.
13. Cao Q, Buskens E, Feenstra T, Jaarsma T, Hillege H, Postmus D. Continuous-time semi-markov models in health economic decision making: an illustrative example in heart failure disease management. Med Decis Making. 2016;36:59–71.
14. Van Rosmalen J, Toy M, O'Mahony JF. A Mathematical approach for evaluating Markov models in continuous time without discrete-event simulation. Med Decis Making. 2013;33:767–79.
15. Sanders GD, Neumann PJ, Basu A, Brock DW, Feeny D, Krahn M, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: second panel on cost-effectiveness in health and medicine. JAMA. 2016;316:1093.
16. Andersson TM-L, Dickman PW, Eloranta S, Lambe M, Lambert PC. Estimating the loss in expectation of life due to cancer using flexible parametric survival models. Stat Med. 2013;32:5286–300.
17. Andersson TM-L, Rutherford MJ, Lambert PC. Illustration of different modelling assumptions for estimation of loss in expectation of life due to cancer. BMC Med Res Methodol. 2019;19:145.
18. Sweeting MJ, Rutherford MJ, Jackson D, Lee S, Latimer NR, Hettle R, et al. Survival extrapolation incorporating general population mortality using excess hazard and cure models: a tutorial. Med Decis Making. 2023;43:737–48.
19. Van Oostrum I, Ouwens M, Remiro-Azócar A, Baio G, Postma MJ, Buskens E, et al. Comparison of parametric survival extrapolation approaches incorporating general population mortality for adequate health technology assessment of new oncology drugs. Value Health. 2021;24:1294–301.
20. Hwang J-S, Hu T-H, Lee LJ-H, Wang J-D. Estimating lifetime medical costs from censored claims data. Health Econ. 2017;26:e332–44.
21. Chen EY-T, Leontyeva Y, Lin C-N, Wang J-D, Clements MS, Dickman PW. Comparing survival extrapolation within all-cause and relative survival frameworks by standard parametric models and flexible parametric spline models using the Swedish Cancer Registry. Med Decis Making. 2024;44:269–82.
22. Manevski D, Putter H, Pohar Perme M, Bonneville EF, Schetelig J, De Wreede LC. Integrating relative survival in multi-state models—a non-parametric approach. Stat Methods Med Res. 2022;31:997–1012.
23. Gilard-Pioc S, Abrahamowicz M, Mahboubi A, Bouvier A-M, Dejardin O, Huszti E, et al. Multi-state relative survival modelling of colorectal cancer progression and mortality. Cancer Epidemiol. 2015;39:447–55.
24. Huszti E, Abrahamowicz M, Alioum A, Binquet C, Quantin C. Relative survival multistate Markov model. Stat Med. 2012;31:269–86.
25. Gillaizeau F, Dantan E, Giral M, Foucher Y. A multistate additive relative survival semi-Markov model. Stat Methods Med Res. 2017;26:1700–11.
26. Latimer N. NICE DSU Technical Support Document 14: undertaking survival analysis for economic evaluations alongside clinical trials—extrapolation with patient-level data [Internet]. 2011. Available from: http://www.nicedsu.org.uk.
27. O'Hagan A, Stevenson M, Madan J. Monte Carlo probabilistic sensitivity analysis for patient level simulation models: efficient estimation of mean and variance using ANOVA. Health Econ. 2007;16:1009–23.
28. Williams C, Lewsey JD, Briggs AH, Mackay DF. Cost-effectiveness analysis in R using a multi-state modeling survival analysis framework: a tutorial. Med Decis Making. 2017;37:340–52.
29. Hinchliffe SR, Scott DA, Lambert PC. Flexible parametric illness-death models. Stand Genomic Sci. 2013;13:759–75.
30. Andersen PK, Keiding N. Multi-state models for event history analysis. Stat Methods Med Res. 2002;11:91–115.
31. Royston P, Lambert PC. Flexible parametric survival analysis using Stata: beyond the cox model. Texas: Stata Press; 2011.
32. Durrleman S, Simon R. Flexible regression models with cubic splines. Stat Med. 1989;8:551–61.

33. Nelson CP, Lambert PC, Squire IB, Jones DR. Flexible parametric models for relative survival, with application in coronary heart disease. Stat Med. 2007;26:5486–98.
34. Bender R, Augustin T, Blettner M. Generating survival times to simulate Cox proportional hazards models. Stat Med. 2005;24:1713–23.
35. Crowther MJ, Lambert PC. Simulating biologically plausible complex survival data. Stat Med. 2013;32:4118–34.
36. Willekens FJ, Putter H. Software for multistate analysis. DemRes. 2014;31:381–420.
37. Jackson CH. Multi-state models for panel data: the **msm** package for *R*. J Stat Soft [Internet]. 2011 [cited 2024 Oct 4];38. Available from: http://www.jstatsoft.org/v38/i08/.
38. Karlsson A, Olofsson N, Laure E, Clements M. A parallel microsimulation package for modelling cancer screening policies. In: 2016 IEEE 12th international conference on e-science (e-Science) [Internet]. Baltimore: IEEE; 2016 [cited 2024 May 16]. p. 323–30. Available from: http://ieeexplore.ieee.org/document/7870915/.
39. Fiocco M, Putter H, Van Houwelingen HC. Reduced-rank proportional hazards regression and simulation-based prediction for multi-state models. Stat Med. 2008;27:4340–58.
40. Hallek M, Fischer K, Fingerle-Rowson G, Fink A, Busch R, Mayer J, et al. Addition of rituximab to fludarabine and cyclophosphamide in patients with chronic lymphocytic leukaemia: a randomised, open-label, phase 3 trial. Lancet. 2010;376:1164–74.
41. National Institute for Health and Clinical Excellence. Rituximab for the first line treatment of chronic lymphocytic leukaemia [Internet]. 2009. Available from: http://guidance.nice.org.uk/TA174.
42. Roche Products Limited. Rituximab for the 1st line treatment of chronic lymphocytic leukaemia [Internet]. National Institute for Health and Clinical Excellence; 2008. Available from: http://www.nice.org.uk/guidance/ta174/resources/roche-submission2.
43. Fischer K, Bahlo J, Fink AM, Goede V, Herling CD, Cramer P, et al. Long-term remissions after FCR chemoimmunotherapy in previously untreated patients with CLL: updated results of the CLL8 trial. Blood. 2016;127:208–15.
44. University of California Berkeley and Max Planck Institute for Demographic Research. The human mortality database [Internet]. Available from: https://www.mortality.org.
45. Guyot P, Ades A, Ouwens MJ, Welton NJ. Enhanced secondary analysis of survival data: reconstructing the data from published Kaplan–Meier survival curves. BMC Med Res Methodol. 2012;12:9.
46. Lambert PC, Royston P. Further development of flexible parametric models for survival analysis. Stand Genomic Sci. 2009;9:265–90.
47. Akaike H. Maximum likelihood identification of Gaussian autoregressive moving average models. Biometrika. 1973;60:255–65.
48. Structured Data LLC. The beta-PERT distribution. Available from: https:// www.riskamp.com/beta-pert.
49. Hess K, Gentleman R. muhaz: hazard function estimation in survival analysis. 2021.
50. Rebora P, Salim A, Reilly M. bshazard: a flexible tool for nonparametric smoothing of the hazard function. R J. 2014;6:114.
51. Jackson C. flexsurv: a platform for parametric survival modeling in R. J Stat Soft [Internet]. 2016 [cited 2024 May 16];70. Available from: http://www.jstatsoft.org/v70/i08/.
52. Liu X-R, Pawitan Y, Clements M. Parametric and penalized generalized survival models. Stat Methods Med Res. 2018;27:1531–46.
53. Couzin-Frankel J. Cancer immunotherapy. Science. 2013;342:1432–3.
54. Sterner RC, Sterner RM. CAR-T cell therapy: current limitations and potential strategies. Blood Cancer J. 2021;11:69.
55. Latimer NR, Rutherford MJ. Mixture and non-mixture cure models for health technology assessment: what you need to know. Pharmacoeconomics. 2024;42:1073–90.
56. Briggs A, Sculpher M. An introduction to Markov modelling for economic evaluation. Pharmacoeconomics. 1998;13:397–409.
57. Eloranta S, Smedby KE, Dickman PW, Andersson TM. Cancer survival statistics for patients and healthcare professionals—a tutorial of real-world data analysis. J Intern Med. 2021;289:12–28.
58. Bright CJ, Brentnall AR, Wooldrage K, Myles J, Sasieni P, Duffy SW. Errors in determination of net survival: cause-specific and relative survival settings. Br J Cancer. 2020;122:1094–101.